

Award Abstract # 2335642

SBIR Phase II: A Hardware-Aware AutoML Platform for Resource-Constrained Devices

NSF Org:	
<u>TI</u> <u>Translational Impacts</u>	
Recipient:	
AI POW LLC	
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Program Manager:	
Peter Atherton patherto@nsf.gov (703)292-8772 TI Translational Impacts TIP Directorate for Technology, Innovation, and Partnerships	Feedback
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History of Investigator:
Alfredo Costilla Reyes (Principal Investigator) alfredo.costilla.reyes@gmail.com
Recipient Sponsored Research Office:
AI POW LLC 2605 SOMERTON CT COLLEGE STATION TX US 77845-7466 (480)265-6388
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10
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AI POW LLC 4321 Jim West St. Bellaire TX US 77401-5201
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ABSTRACT

The broader impact of this Small Business Innovation Research (SBIR) Phase II project will support industries keen on harnessing the power of artificial intelligence and Internet of Things. By simplifying the deployment of artificial intelligence technologies, this project allows a broader range of businesses in the manufacturing sector to join the data revolution. It empowers businesses to leverage their existing data assets, leading to enhanced efficiency, fostering a culture of innovation, and carving out a competitive edge in the market. On the commercial front, the benefits are multifold, ranging from bolstered business efficiency and substantial cost reductions to potential market expansion for solutions in artificial intelligence of things. From a societal perspective, this technology contributes substantially to the development of a data-literate 21st-century workforce and strengthens human-technology synergies. The project will drive efficiency and standardization across diverse industries and streamline the process of analyzing and acting upon extensive data sets which will result in improved product quality, fuels

innovation, and pave the way for more efficient decision-making processes in an increasingly data-driven world.

This Small Business Innovation Research Phase II project addresses the complex challenge of efficiently deploying artificial intelligence models on edge devices for real-time defect detection in industrial manufacturing systems. The problem lies in creating a scalable, efficient, and easy-to-use solution that allows for the wide application of artificial intelligence technologies in Internet of Things devices. The research objectives include developing a modular end-to-end defect detection system, implementing advanced machine learning automation techniques, improving model interpretability, and enhancing model compression for edge devices. The research will leverage machine learning, edge computing, and user feedback to create a practical, robust, and user-friendly solution. The anticipated results include an artificial intelligence of things system that effectively performs real-time defect detection with improved interpretability and reduced resource usage. The expected outcomes comprise an artificial intelligence of things system capable of performing real-time defect detection with elevated interpretability, using minimal computational resources, and revolutionizing defect detection in industrial manufacturing systems.

This award reflects NSF's statutory mission and has been deemed worthy of support through evaluation using the Foundation's intellectual merit and broader impacts review criteria.

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Sui, Songyuan and Deng, Zhanqi and Zhuang, Yamin and Liu, Serena and Yu, Leisheng and Costilla_Reyes, Alfredo and Hu, Xia "ECLAD: An Edge-Cloud Collaborative Agentic Framework for Interpretable Anomaly Detection in Predictive Maintenance" , 2026 <https://doi.org/10.1145/3774905.3793127> [Citation Details](#)

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